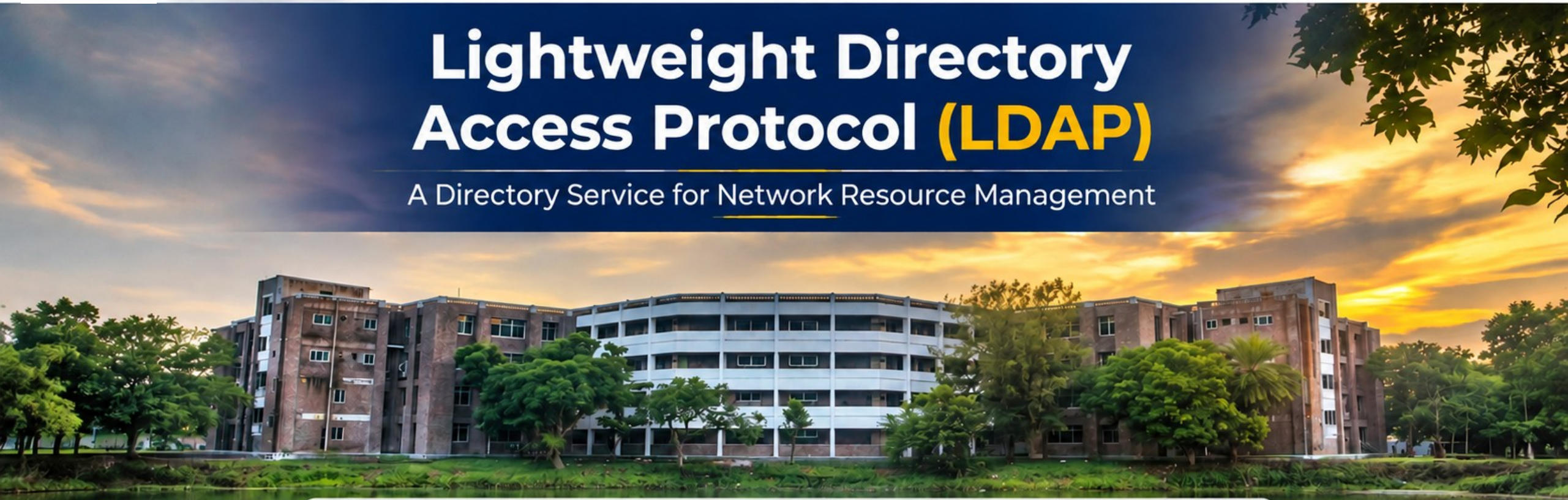
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Lightweight Directory Access Protocol (LDAP)

A Directory Service for Network Resource Management



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Course: ICT-3101
Course Name: **Data Communication**

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Presentation outline

01	**LDAP fundamentals**	What LDAP is and why directory services are useful
02	**Directory model**	Entries, attributes, DIT and Distinguished Names
03	**Architecture & workflow**	Client–server communication and LDAP operations
04	**Security & deployment**	Authentication, TLS, access control and common uses
05	**Key takeaways**	Strengths, limitations and references

What is LDAP?

LDAP stands for Lightweight Directory Access Protocol.

It is an open, standards-based application-layer protocol used by clients to access and manage information held by a directory service.

Protocol, not a database

LDAP defines how a client communicates with a directory service. The service stores and organizes the directory data.

Optimized for lookup

Directory systems are designed for frequent reads and searches, such as finding users, groups or devices.

Standardized interface

Different directory products can expose a common LDAP interface, although their features may vary.

LDAP version 3 is the widely deployed version

Why do we use LDAP?

Organizations often need one consistent place to look up identity and resource information.

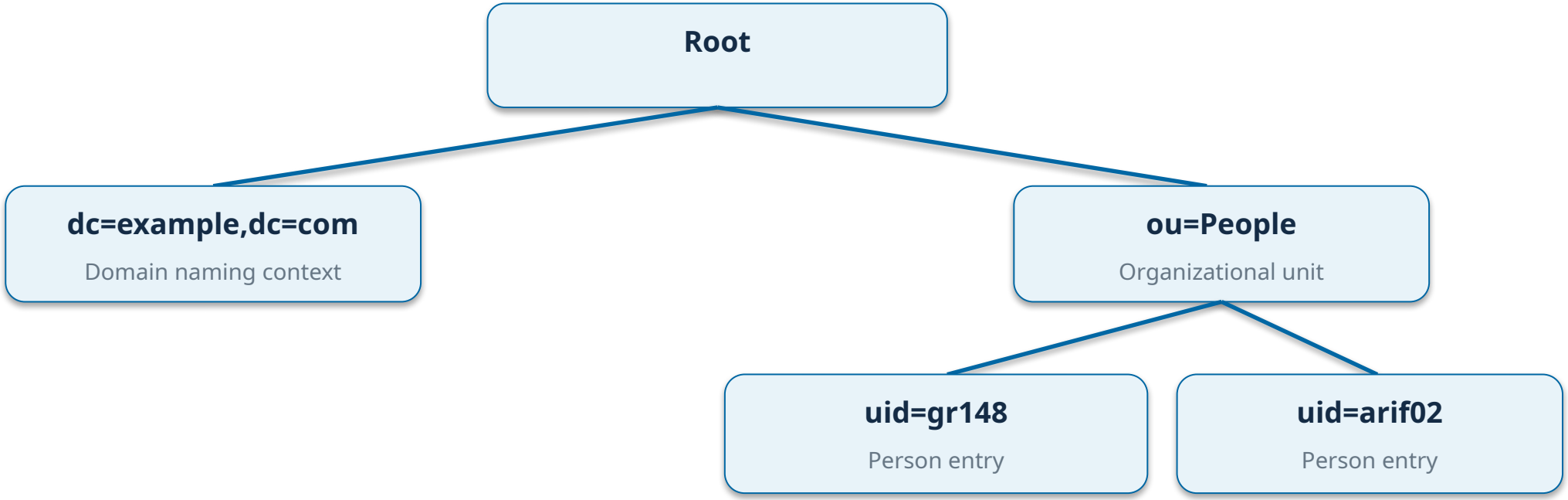
01	**Centralized directory**	Store user, group, device and organizational records in a stru
02	**Consistent lookup**	Applications can search the same directory instead of maintaining
03	**Access management**	Directory groups and attributes can help applications make a
04	**Scalability**	Hierarchical naming and indexing support large directory trees an

Example

A university can use a directory to look up a student's account, department, email address and group membership.

How directory information is organized

LDAP directories commonly represent information as a Directory Information Tree (DIT).



Each node is an entry. An entry contains attributes such as cn, sn, mail, uid and objectClass.

Entries, attributes and Distinguished Names

Entry

A directory record representing an object, such as a person, group, device or organizational unit.

Attribute

A named property of an entry. Examples: cn (common name), sn (surname), mail and uid.

DN

The Distinguished Name uniquely identifies an entry within the directory information tree.

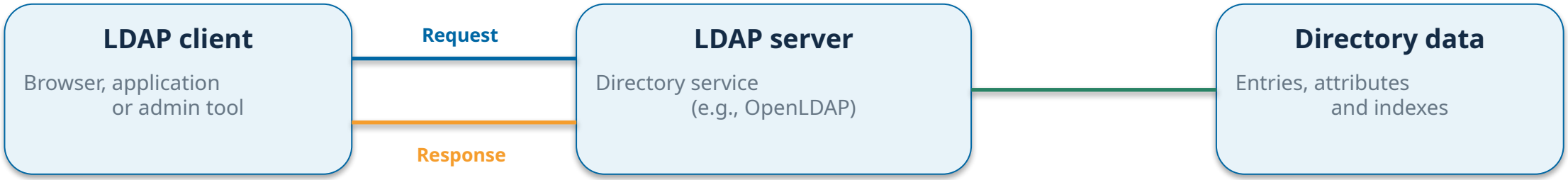
Example entry

```
dn: uid=gr148,ou=People,dc=example,dc=com
objectClass: inetOrgPerson
uid: gr148
cn: Golam Rabby
sn: Rabby
mail: gr148@example.com
```

The DN is read from the specific entry outward through its parent components.

LDAP client–server architecture

LDAP follows a request–response model over a network connection.



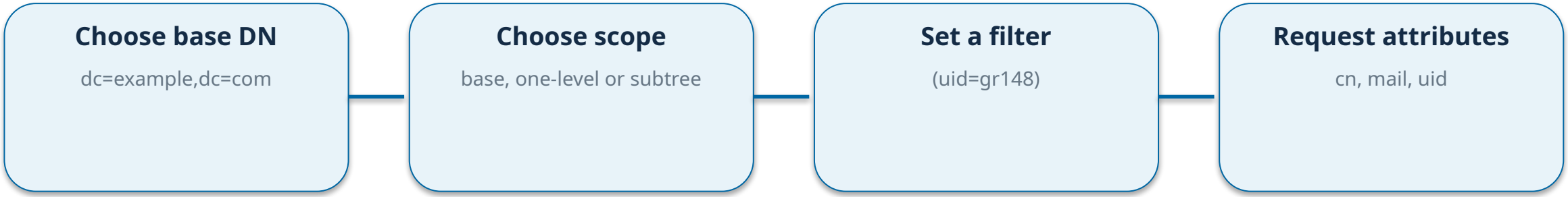
The server evaluates the request, applies access rules, reads or updates directory data, and returns a result.

LDAP: application layer **Usually over TCP**

Core LDAP operations

01	**Bind**	Authenticate or establish the LDAP session context.
02	**Search**	Find entries using a base DN, scope and filter.
03	**Add**	Create a new directory entry.
04	**Modify**	Change attribute values in an existing entry.
05	**Delete**	Remove an entry from the directory.
06	**Modify DN**	Rename an entry or move it within the directory tree.
07	**Compare**	Check whether an entry has a specified attribute value.
08	**Unbind**	End the LDAP session.

How an LDAP search works



Illustrative LDAP search filter

```
(&(objectClass=inetOrgPerson)(uid=gr148))
```

This filter asks for entries that are both inetOrgPerson objects and have uid equal to gr148.

A search result returns matching entries and the requested attributes, subject to access controls.

LDAP transport and security

TCP port 389

Common LDAP port. It can carry LDAP traffic and can be protected with StartTLS.

TCP port 636

Commonly used for LDAP over TLS from the start of the connection (LDAPS).

TLS protection

Encrypts traffic and helps protect credentials and directory data in transit.

Security best practices

- Use TLS (StartTLS or LDAPS) when transmitting credentials or sensitive data.
- Do not send simple-bind passwords over an unencrypted connection.
- Apply least-privilege access controls to directory entries and attributes.
- Use strong authentication, monitor access and keep directory software updated.

LDAP and related technologies

Technology	What it is	How it relates to LDAP
LDAP	A directory access protocol	Defines how clients query and modify directory entries.
OpenLDAP	An LDAP directory software implementation	Provides an LDAP server and related tools.
Microsoft Active Directory	A directory service for Windows environments	Supports LDAP, and also uses other protocols and services such as Kerberos.
SQL database	A relational database system	Uses tables and SQL; it is not an LDAP directory protocol.

Remember: LDAP is the protocol; OpenLDAP and Active Directory are directory-service products.

Common applications of LDAP

Single sign-on foundations

Applications can consult a shared directory for identity information; SSO typically also requires an authentication/federation mechanism.

User and group lookup

Find account details, group membership, department and contact information.

Enterprise address books

Search staff names, email addresses, phone numbers and organizational units.

Device and service configuration

Store structured configuration or network-resource information where a directory model is appropriate.

Example environment: a university directory can organize students, teachers, departments, groups and email contacts.

Key takeaways

01 **LDAP is an application-layer protocol for accessing and managing directory information.**

02 Directory data is organized as entries and attributes in a hierarchical Directory Information Tree.

03 A Distinguished Name identifies an entry; search operations locate entries using a base, scope and filter.

04 Bind, Search, Add, Modify and Delete are among the core LDAP operations.

05 Use TLS, strong authentication and least-privilege access controls to secure directory communications.

In short: LDAP provides a common, structured way for networked applications to work with directory services.

Selected references

- [1] M. Wahl, T. Howes and S. Kille, “Lightweight Directory Access Protocol (v3),” RFC 4511, IETF, 2006.
- [2] K. Zeilenga, “Lightweight Directory Access Protocol (LDAP): Directory Information Models,” RFC 4512, IETF, 2006.
- [3] R. Harrison, “Lightweight Directory Access Protocol (LDAP): Authentication Methods and Security Mechanisms,” RFC 4513, IETF, 2006.
- [4] OpenLDAP Project, OpenLDAP Administrator’s Guide.
- [5] Microsoft Learn, “Active Directory Domain Services overview” and LDAP-related documentation.

The RFC documents define LDAPv3 protocol behaviour, directory information models and security mechanisms.

Thank you

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